GUARDIAN ANGEL

Problem Statement / Rationale

All around the world, many people who live in poverty suffer a myriad of illnesses. With a lack of access to money, education, or a reliable healthcare system, these people resort to home remedies instead of seeking professional medical attention. In places with weak or no signal at all, it becomes difficult to research sicknesses or medicine suitable for them. On top of that, unclean surroundings create the perfect breeding ground for pathogens and bacteria to spread. In such an environment, the chances for an epidemic to start within the area increase.

Project Overview and Relevance

Guardian Angel can be programmed as an application for laptops/personal computers or for mobile devices. It will analyze the user's symptoms and provide an easy and more accurate diagnosis for a simple illness, but the app will encourage the user to visit a doctor if the symptoms resemble something much deeper. This will help not just people living in poverty but also retired senior citizens, young children, and everyone else in between.

In addition to diagnosing the patient, it also calculates their BMI. It will also use and keep track of laboratory results, such as blood chemistry. It can use these values to warn the patient of potential health risks, such as high cholesterol levels and uric acid, both of which can cause damage to major organs and bones.

What follows is the list of modules the program is divided into:

- **Log-in/Sign-up**: Allows users to create an account or access their existing profile.
- **User Profile**: Collects basic health information (age, weight, height, etc.)
- **Health Data Input**: Enables users to input their daily steps, calories, water intake, workouts, and other relevant details.

- **Diet Tracker**: Tracks the diet you list down, offers guidance on what to eat next to maintain a healthy lifestyle, what to keep & stop eating (according to the foods' nutritional value), and explains why these foods could be harmful.

- **Disease Determiner/Advice Provider**: Determines the disease based on symptoms, and gives the user advice on how to treat it based on reliable information online, like consuming medication often available at home.

- **Reliability verifier:** Decides whether the information online used by the Advice Provider is accurate, based on criteria, such as the quality of the explanation backing it, and its website/source.

- **Health Monitor**: Tracks heart rate, blood pressure, blood sugar, and other values, warns the user of potential danger when they approach a critical level, and provides tips on how to avoid that danger by lowering the mentioned values.

- **Disease Significance Estimator**: Identifies the seriousness of the disease, as determined by the Disease Determiner, and advises the user to seek medical advice from professionals if the condition is too severe.

Objectives

- Give the user tips on how to cure most diseases
- Provides the user, regardless of age, with information about hospital laboratory results
- Provide the user with the healing capabilities nearly comparable to those of medical facilities at the comfort of their homes (with limitations like severe injuries, unavailability of medicine, inaccuracy of diagnoses)
- Offer the user accurate, consistent information regarding diseases and methods for their treatment, without putting them through the trouble of visiting several websites
- Deliver immediate, quality responses to users' queries about illnesses

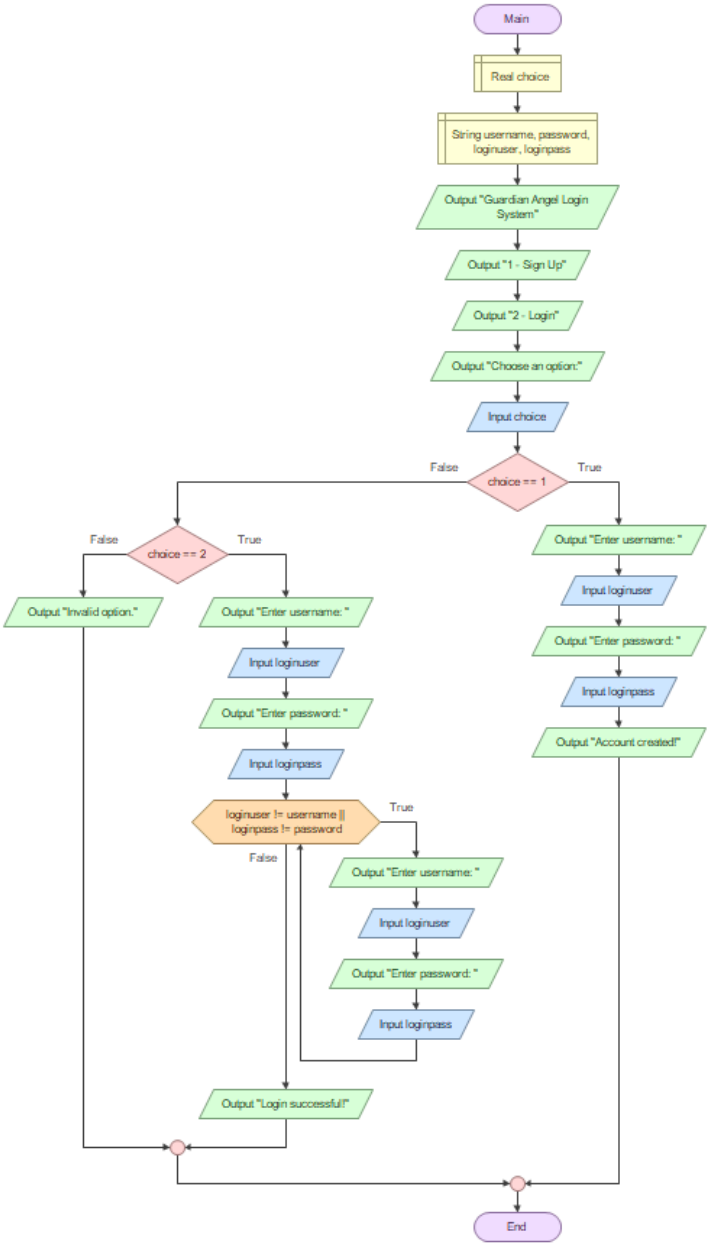
Methodology / Technical Approach

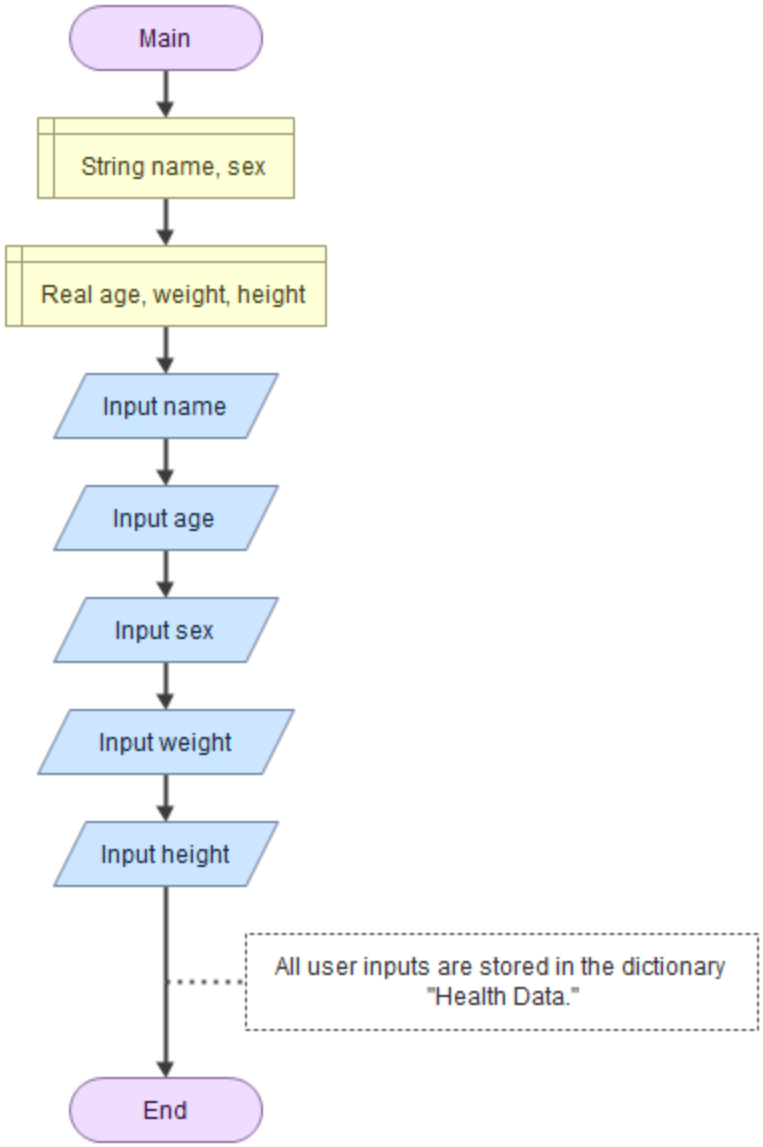
Before the main goal and ideas were identified, the technical approach was first discussed. The language the Guardian Angel program uses is Python because of its significant practicality and efficiency compared to other programming languages, such as C++, JSON/Flask, etc. Then, we implemented the ideas and program structure, which composed the first versions of Guardian Angel.

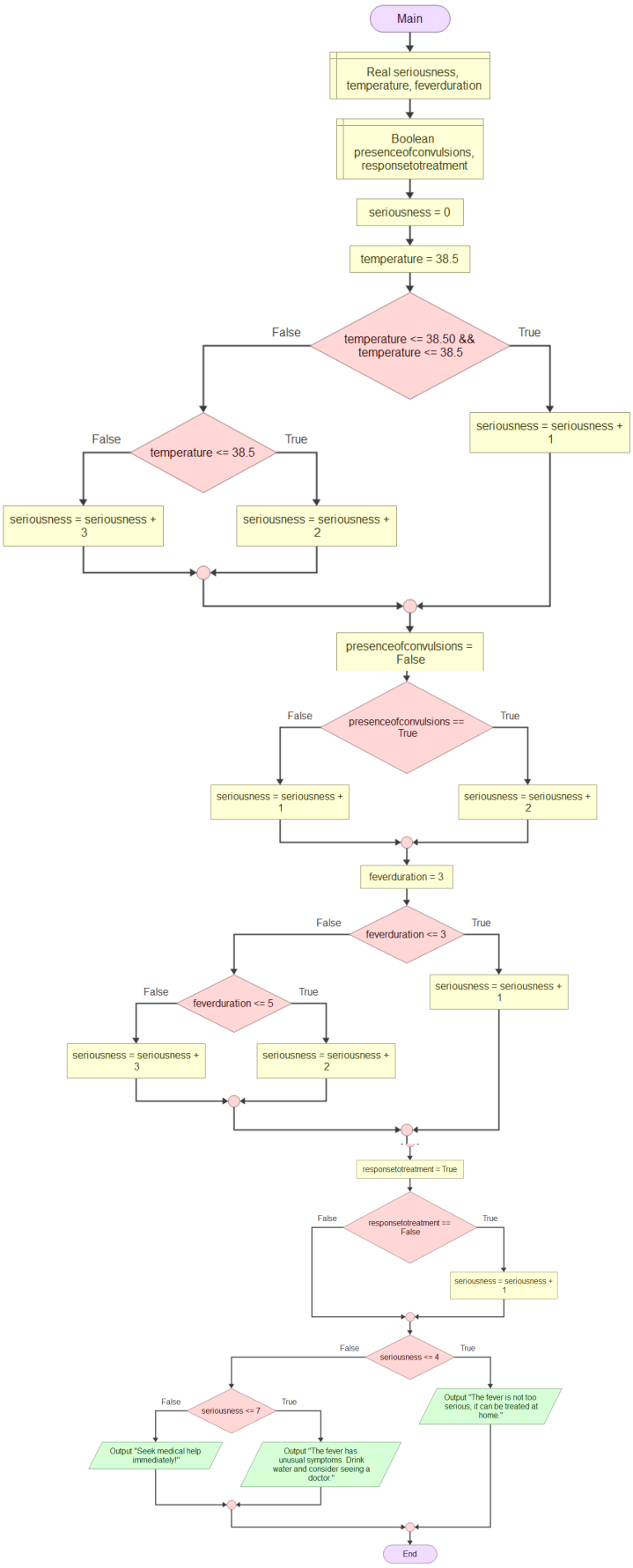
During the earlier versions, it was messy, unworkable, and full of bugs, due to overlapping variables and values, causing the data to mix up and ruin the program. However, as the program was developed further, the errors were managed and fixed, making the program usable overall.

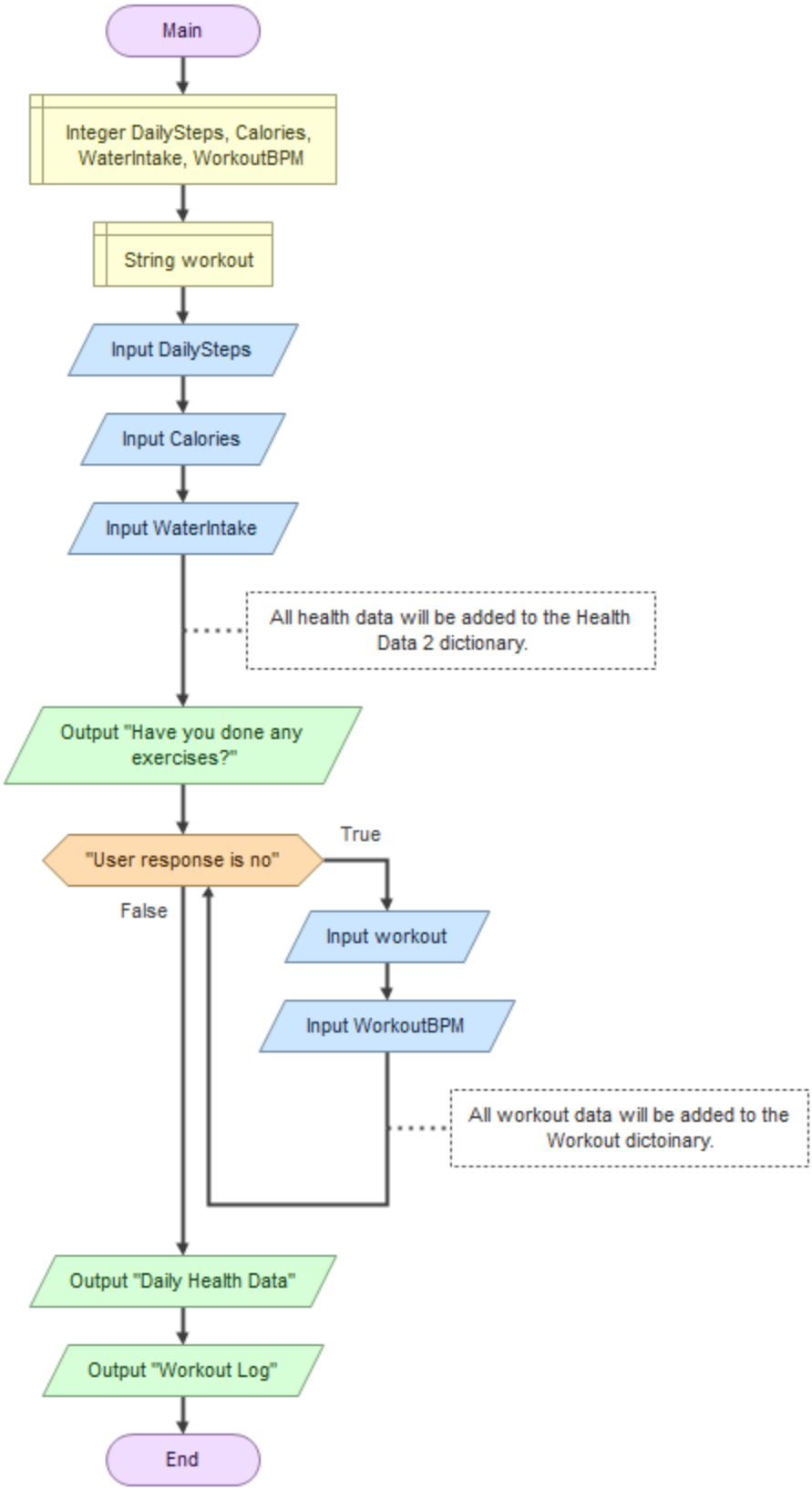
Constructing the project was a very interesting and thrilling experience. We worked on it for days and weeks. It even offered experiences of fun and enjoyable times together, which united the group in order to make the best program to satisfy and serve the public.

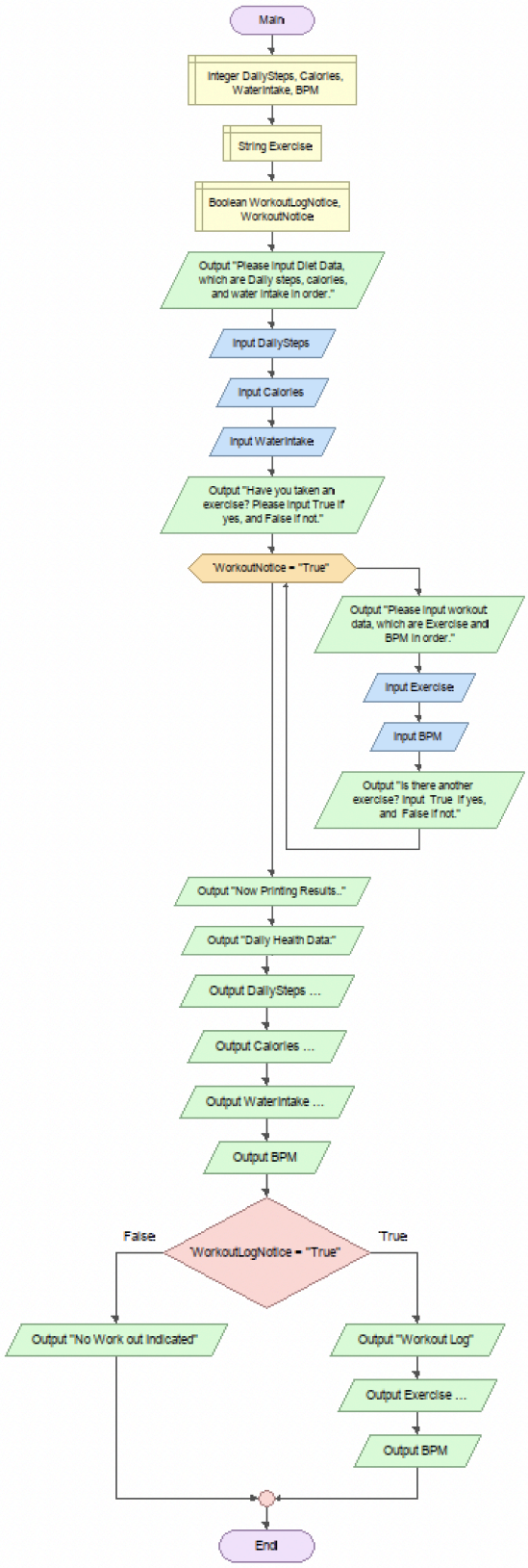
Flowchart

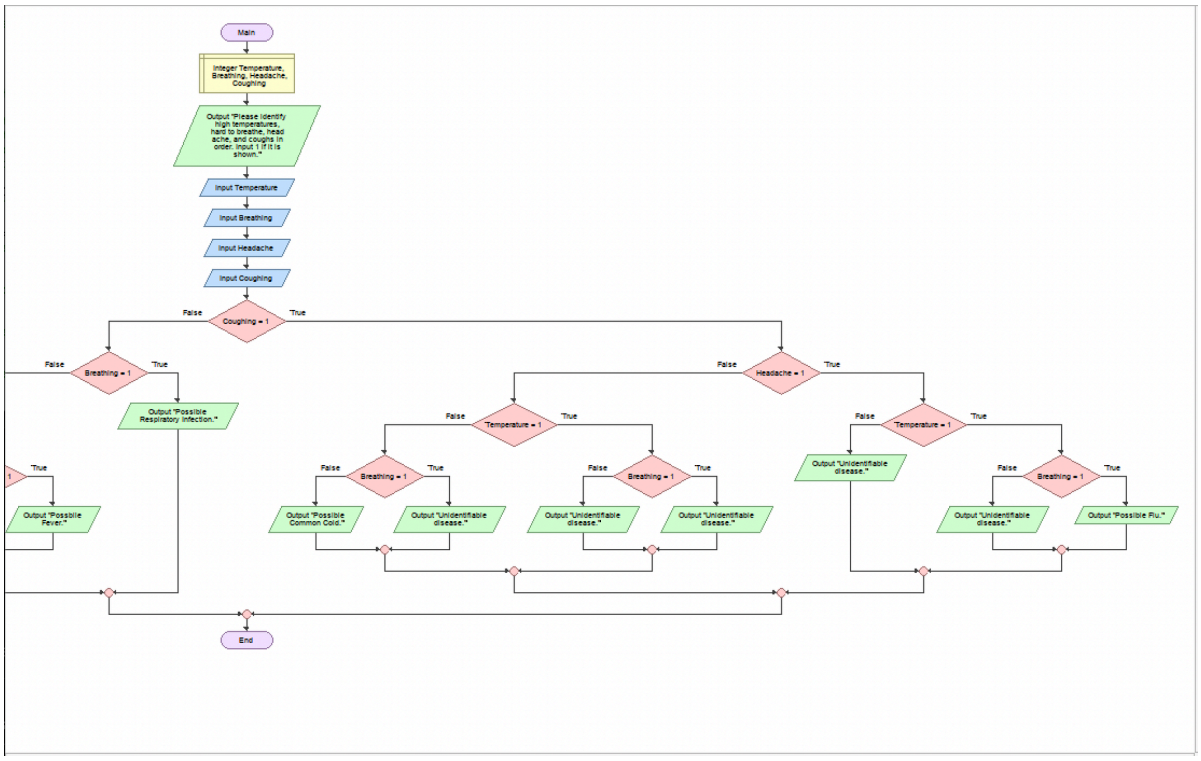

Main
Real choice
String username, password, loginuser, loginpass
Output "Guardian Angel Login System"
Output "1 - Sign Up"
Output "2 - Login"
Output "Choose an option:"
Input choice
choice == 1
False
True
Output "Enter username: "
Input loginuser
Output "Enter password: "
Input loginpass
Output "Account created!"
choice == 2
False
True
Output "Invalid option."
Output "Enter username: "
Input loginuser
Output "Enter password: "
Input loginpass
loginuser != username || loginpass != password
True
False
Output "Enter username: "
Input loginuser
Output "Enter password: "
Input loginpass
Output "Login successful!"
End

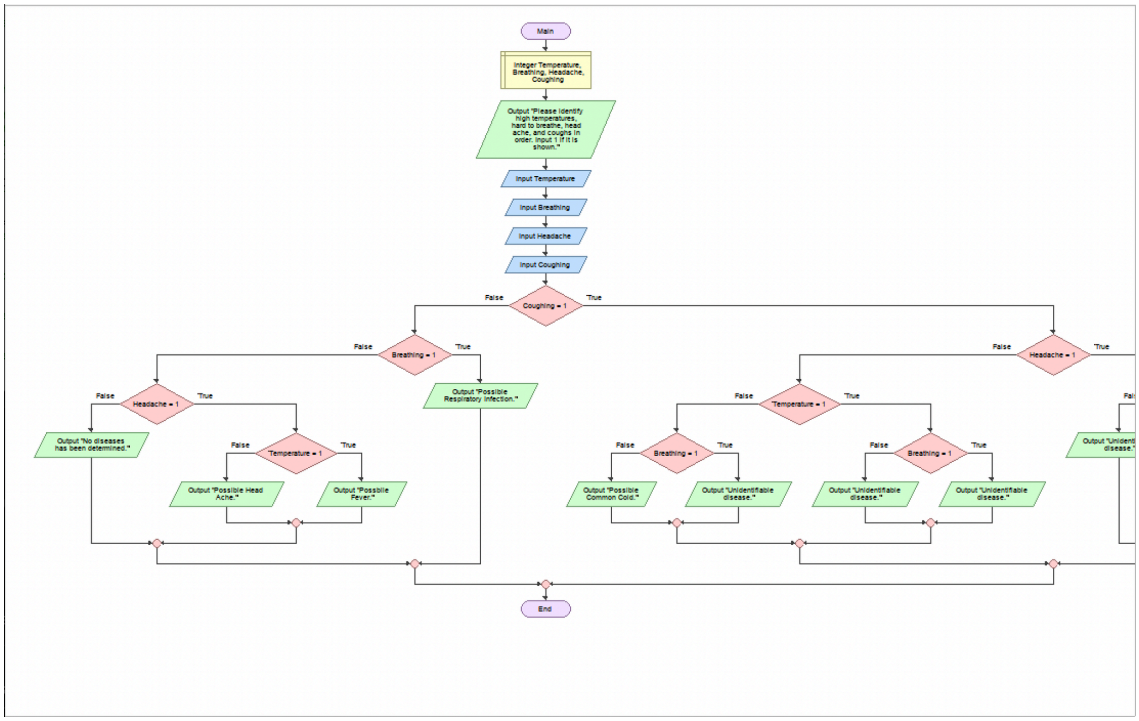

Main
String name, sex
Real age, weight, height
Input name
Input age
Input sex
Input weight
Input height
All user inputs are stored in the dictionary "Health Data."
End

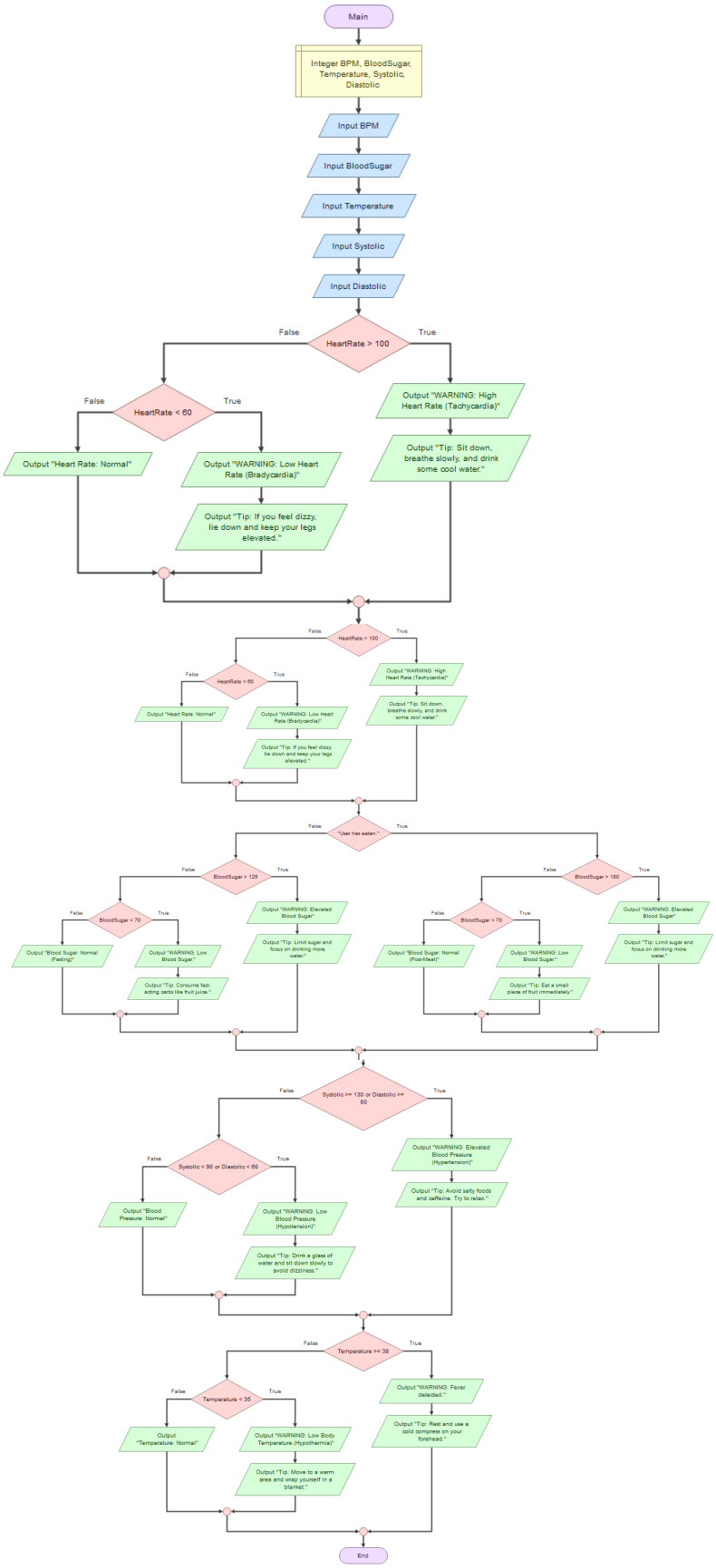

Main
Real seriousness, temperature, feverduration
Boolean presenceofconvulsions, responsetotreatment
seriousness = 0
temperature = 38.5
temperature <= 38.50 && temperature <= 38.5
False
True
seriousness = seriousness + 1
temperature <= 38.5
False
True
seriousness = seriousness + 3
seriousness = seriousness + 2
presenceofconvulsions = False
presenceofconvulsions == True
False
True
seriousness = seriousness + 1
seriousness = seriousness + 2
feverduration = 3
feverduration <= 3
False
True
seriousness = seriousness + 1
feverduration <= 5
False
True
seriousness = seriousness + 3
seriousness = seriousness + 2
responsetotreatment = True
responsetotreatment == False
False
True
seriousness = seriousness + 1
seriousness <= 4
False
True
Output "The fever is not too serious, it can be treated at home."
seriousness <= 7
False
True
Output "Seek medical help immediately!"
Output "The fever has unusual symptoms. Drink water and consider seeing a doctor."
End


Main
Integer DailySteps, Calories, WaterIntake, WorkoutBPM
String workout
Input DailySteps
Input Calories
Input WaterIntake
All health data will be added to the Health Data 2 dictionary.
Output "Have you done any exercises?"
"User response is no"
True
False
Input workout
Input WorkoutBPM
All workout data will be added to the Workout dictoinary.
Output "Daily Health Data"
Output "Workout Log"
End


Main
Integer DailySteps, Calories, WaterIntake, BPM
String Exercise
Boolean WorkoutLogNotice, WorkoutNotice
Output "Please Input Diet Data, which are Daily steps, calories, and water intake in order."
Input DailySteps
Input Calories
Input WaterIntake
Output "Have you taken an exercise? Please Input True if yes, and False if not."
WorkoutNotice = "True"
Output "Please Input workout data, which are Exercise and BPM in order."
Input Exercise
Input BPM
Output "Is there another exercise? Input True if yes, and False if not."
Output "Now Printing Results.."
Output "Daily Health Data:"
Output DailySteps ...
Output Calories ...
Output WaterIntake ...
Output BPM
False
WorkoutLogNotice = "True"
True
Output "No Work out indicated"
Output "Workout Log"
Output Exercise ...
Output BPM
End


Main
Integer Temperature, Breathing, Headache, Coughing
Output "Please identify high temperatures, hard to breathe, head ache, and coughs in order. Input 1 if it is shown."
Input Temperature
Input Breathing
Input Headache
Input Coughing
Coughing = 1
False
True
Breathing = 1
False
True
Output "Possible Respiratory Infection."
Headache = 1
False
True
Temperature = 1
False
True
Breathing = 1
False
True
Output "Possible Common Cold."
Output "Unidentifiable disease."
Breathing = 1
False
True
Output "Unidentifiable disease."
Output "Unidentifiable disease."
Temperature = 1
False
True
Output "Unidentifiable disease."
Breathing = 1
False
True
Output "Unidentifiable disease."
Output "Possible Flu."
Output "Possible Fever."
End


Main
Integer Temperature, Breathing, Headache, Coughing
Output "Please identify high temperatures, hard to breathe, head ache, and coughs in order. Input 1 if it is shown."
Input Temperature
Input Breathing
Input Headache
Input Coughing
Coughing = 1
False
True
Breathing = 1
False
True
Output "Possible Respiratory Infection."
Headache = 1
False
True
Output "No diseases has been determined."
Temperature = 1
False
True
Output "Possible Head Ache."
Output "Possible Fever."
Headache = 1
False
True
Temperature = 1
False
True
Breathing = 1
False
True
Output "Possible Common Cold."
Output "Unidentifiable disease."
Breathing = 1
False
True
Output "Unidentifiable disease."
Output "Unidentifiable disease."
End


Main
Integer BPM, BloodSugar, Temperature, Systolic, Diastolic
Input BPM
Input BloodSugar
Input Temperature
Input Systolic
Input Diastolic
HeartRate > 100
False
True
HeartRate < 60
Output "Heart Rate: Normal"
Output "WARNING: Low Heart Rate (Bradycardia)"
Output "Tip: If you feel dizzy, lie down and keep your legs elevated."
Output "WARNING: High Heart Rate (Tachycardia)"
Output "Tip: Sit down, breathe slowly, and drink some cool water."
"User has eaten."
BloodSugar > 125
BloodSugar < 70
Output "Blood Sugar: Normal (Fasting)"
Output "WARNING: Low Blood Sugar."
Output "Tip: Consume fast-acting carbs like fruit juice."
Output "WARNING: Elevated Blood Sugar"
Output "Tip: Limit sugar and focus on drinking more water."
BloodSugar > 180
BloodSugar < 70
Output "Blood Sugar: Normal (Post-Meal)"
Output "WARNING: Low Blood Sugar."
Output "Tip: Eat a small piece of fruit immediately."
Output "WARNING: Elevated Blood Sugar"
Output "Tip: Limit sugar and focus on drinking more water."
Systolic >= 130 or Diastolic >= 80
Systolic < 90 or Diastolic < 60
Output "Blood Pressure: Normal"
Output "WARNING: Low Blood Pressure (Hypotension)"
Output "Tip: Drink a glass of water and sit down slowly to avoid dizziness."
Output "WARNING: Elevated Blood Pressure (Hypertension)"
Output "Tip: Avoid salty foods and caffeine. Try to relax."
Temperature >= 38
Temperature < 35
Output "Temperature: Normal"
Output "WARNING: Low Body Temperature (Hypothermia)"
Output "Tip: Move to a warm area and wrap yourself in a blanket."
Output "WARNING: Fever detected."
Output "Tip: Rest and use a cold compress on your forehead."
End

Program

```
Guardian Angel Login System
1 - Sign Up
2 - Login
Choose an option: 2
Enter username: username
Enter password: password
Login successful!
Enter your name: Gian
Enter your age: 13
Enter your sex: Male
Enter your weight (kg): 56
Enter your height (cm): 166

======== GUARDIAN ANGEL MENU ========

1 - Diet Tracker
2 - Disease Determiner (with Reliability Verifier!)
    2.1 - Disease Significance Estimator
3 - Health Monitor (coming soon)

Select feature: 1
Input Daily Steps: 1000
Input Calories: 2000
Input Water Intake: 400
Have you done any exercises? Yes/No: Yes
Enter workout: Jogging
Enter BPM for that workout: 110
Would you like to continue? Enter No if not: No
Daily Health Data
Daily_Steps: 1000
Calories: 2000
Water_Intake: 400
Workout Log
Jogging: 110 BPM
Return to menu? Yes/No: Yes

======== GUARDIAN ANGEL MENU ========

1 - Diet Tracker
2 - Disease Determiner (with Reliability Verifier!)
    2.1 - Disease Significance Estimator
3 - Health Monitor (coming soon)

Select feature: 2
```

```
======== GUARDIAN ANGEL MENU ========

1 - Diet Tracker
2 - Disease Determiner (with Reliability Verifier!)
    2.1 - Disease Significance Estimator
3 - Health Monitor (coming soon)

Select feature: 2
Between Headache, coughs, high temperature, and hard to breathe.
High temperature? Enter 1 if yes: 1
Hard to breathe? Enter 1 if yes: 1
Headaches? Enter 1 if yes: 1
Coughing? Enter 1 if yes: 1
Possible Flu

Source evaluation: Accurate
Return to menu? Yes/No: Yes

======== GUARDIAN ANGEL MENU ========

1 - Diet Tracker
2 - Disease Determiner (with Reliability Verifier!)
    2.1 - Disease Significance Estimator
3 - Health Monitor (coming soon)

Select feature: 2.1
The fever is not too serious, it can be treated at home.
Return to menu? Yes/No: Yes

======== GUARDIAN ANGEL MENU ========

1 - Diet Tracker
2 - Disease Determiner (with Reliability Verifier!)
    2.1 - Disease Significance Estimator
3 - Health Monitor (coming soon)

Select feature: 2
Between Headache, coughs, high temperature, and hard to breathe.
High temperature? Enter 1 if yes: 1
Hard to breathe? Enter 1 if yes: 1
Headaches? Enter 1 if yes: 0
Coughing? Enter 1 if yes: 0
Possible Respiratory Infection

Source evaluation: Accurate
Return to menu? Yes/No: No
```

Team Roles

Members	Roles
Anish Dela Cruz	Programming assistant, project paper contributor, idea developer
Zoe Ibarra	Programming assistant, project defense presentation creator, idea developer
Marc Aeron Magallanes	Programmer, major contributor to project paper, idea developer
Gian Medroso	Programmer, project paper contributor, idea developer

References / Background Research

World Health Organization: WHO. (2026, January 26). *Healthy diet*.

https://www.who.int/news-room/fact-sheets/detail/healthy-diet

Built-in functions. (n.d.). Python Documentation.

https://docs.python.org/3/library/functions.html

README. File

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Guardian Angel

Guardian angel fetches its user's health data, and determines their physical health using the data given.

Objectives

- Help the person maintain a health diet
- Help the person adjust their lifestyle to be healthy
- Identify whether or not they have a disease, and detect what it is

Features

- Health Diet Tracker
- Disease Determiner
- Reliability Verifier (Addition to the Disease Determiner)
- Health Monitor

Prerequisites

IDLE 3.13.3 or Standard Python

Installation and running of Program

1. Install Python
2. Download Program/Code from Github

Sample Output

```
======== GUARDIAN ANGEL MENU ========

1 - Diet Tracker
2 - Disease Determiner (with Reliability Verifier!)
    2.1 - Disease Significance Estimator
3 - Health Monitor (coming soon)

Select feature: 2
Between Headache, coughs, high temperature, and hard to breathe.
High temperature? Enter 1 if yes: 1
Hard to breathe? Enter 1 if yes: 1
Headaches? Enter 1 if yes: 1
Coughing? Enter 1 if yes: 1
Possible Flu

Source evaluation: Accurate
Return to menu? Yes/No: Yes

======== GUARDIAN ANGEL MENU ========

1 - Diet Tracker
2 - Disease Determiner (with Reliability Verifier!)
    2.1 - Disease Significance Estimator
3 - Health Monitor (coming soon)

Select feature: 2.1
The fever is not too serious, it can be treated at home.
Return to menu? Yes/No: Yes

======== GUARDIAN ANGEL MENU ========

1 - Diet Tracker
2 - Disease Determiner (with Reliability Verifier!)
    2.1 - Disease Significance Estimator
3 - Health Monitor (coming soon)

Select feature: 2
Between Headache, coughs, high temperature, and hard to breathe.
High temperature? Enter 1 if yes: 1
Hard to breathe? Enter 1 if yes: 1
Headaches? Enter 1 if yes: 0
Coughing? Enter 1 if yes: 0
Possible Respiratory Infection

Source evaluation: Accurate
Return to menu? Yes/No: No
```

Contributors

Anish Dela Cruz

Zoe Ibarra

Aeron Magallanes

Gian Medroso

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END OF GUARDIAN ANGEL PROJECT PAPER

GROUP 6 - CS 2 - 8 Adelfa